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Development and Evaluation of a Genetic Algorithm for Placental Gene Selection in Predicting Retinopathy of Prematurity

BACHELOR THESIS

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Abstract

Retinopathy of prematurity (ROP) is one of the leading causes of blindness in preterm infants. This thesis examines whether a genetic algorithm can select a small, predictive set of genes from the placental transcriptome, and how reliable that selection is on a dataset of only 30 samples (14 with ROP, 16 without) with 60,664 genes per sample. After normalization and differential expression analysis, the genetic algorithm selects a gene subset from a candidate pool of 124 genes, and performance is estimated with nested cross-validation to avoid *selection bias*. The genetic algorithm reaches a mean AUC of 0.919 ± 0.068, higher than the Random Forest baseline, Ridge, LASSO and elastic-net regression, while using far fewer genes. Additional checks (a permutation test, excluding genes taken from the literature, and replacing the DEG method) show that the performance estimate is not strongly sensitive to these choices, while the identity of individually selected genes is unstable. In the main configuration, the gene HELLPAR, a long non-coding RNA active in the placental trophoblast, was selected in all 10 independent runs and has the largest SHAP contribution, but it is interpreted as a hypothesis, not as a confirmed biomarker. The thesis is a proof of concept for the methodological approach; its conclusions are limited by the small sample size and require validation on an independent cohort.

Keywords: genetic algorithm, feature selection, retinopathy of prematurity, placenta, nested cross-validation, SHAP

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1

Introduction

1.1 Retinopathy of Prematurity: Clinical Picture and Mechanism

Retinopathy of prematurity (ROP) is a retinal disease that affects preterm infants and is one of the leading causes of childhood blindness [1]. The disease develops in two phases [2]. In the first phase, the preterm infant moves from the low-oxygen environment of the uterus to an environment with much more oxygen (hyperoxia, an excess of oxygen in the tissues, often increased further by supplemental oxygen therapy). This change lowers the levels of the growth factors VEGF (vascular endothelial growth factor, a signal that drives blood vessel growth) and IGF-1 (insulin-like growth factor 1), so normal growth of the retinal blood vessels stops. In the second phase, the part of the retina left without blood vessels suffers from oxygen deficiency (hypoxia) and responds by increasing VEGF secretion. This triggers excessive, disorganized growth of new, pathological blood vessels, a process that can lead to retinal detachment and permanent vision loss.

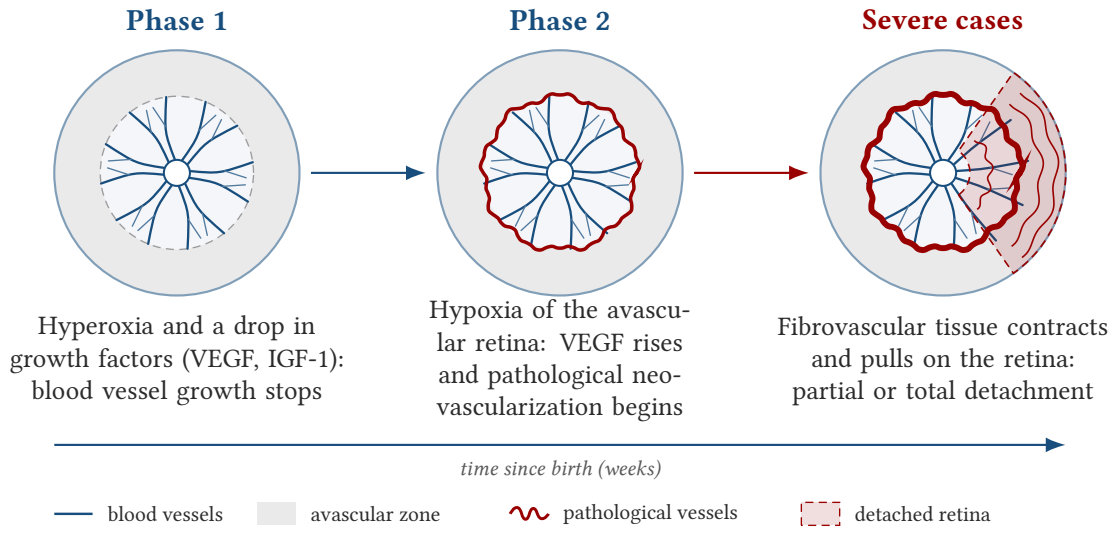


Figure 1.1: Simplified diagram of the two-phase pathogenesis of ROP and the possible outcome in severe cases, based on Hellström et al. [2]. The circles represent the retina viewed from the front, with the optic disc in the center.

Clinical assessment of disease severity uses a standardized international classification,

whose current third edition was published in 2021 [3]. Under current guidelines, screening is performed in infants born with a birth weight below approximately 1500 grams or a gestational age of 30 weeks or less. Beyond these two basic criteria, many additional factors affect the likelihood and severity of the disease. This leaves room to study additional, early predictors of risk. One such direction, involving the role of the placenta, is described in the next section.

1.2 The Role of the Placenta in ROP Risk

The placenta is the organ that regulates the exchange of oxygen and nutrients between mother and fetus during pregnancy. The hypothesis is that the state of the placenta affects how resilient the infant's vascular system, including the retina, will be to the postnatal conditions that lead to ROP [4].

Owen et al. [1] confirmed this link empirically. They showed that placental inflammation in the mother (an inflammatory change of placental tissue, most often in response to infection of the fetal membranes) significantly reduces the child's risk of ROP. Building on that finding, Huang et al. [5] analyzed publicly available placental gene expression data and identified four genes (*MMP25*, *CAPG*, *MCAM*, *B2M*) whose activity is associated with ROP development, and used them to build a risk prediction model.

These findings suggest that placental gene expression carries a measurable signal about ROP risk. This thesis follows the same direction, but instead of standard statistical methods for gene selection it uses a genetic algorithm, described in Chapter 3.

1.3 Machine Learning in ROP Prediction

Beyond clinical assessment, several algorithmic approaches have been developed for early prediction of ROP risk from clinical and laboratory parameters. Table 1.1 summarizes the three most commonly used approaches.

Machine learning is also increasingly applied directly to genomic data. Chen et al. [9] used SVM-RFE, LASSO and Random Forest to identify genes relevant to the diagnosis of ROP associated with sepsis, while Liu and Zhang [10] applied similar methods to placental transcriptomics to find genes shared between gestational diabetes and ROP. None of these studies uses evolutionary algorithms for feature selection or includes an explicit model explainability component (e.g. SHAP). This gap is where this thesis aims to contribute, as discussed in more detail in Chapter 2.

Table 1.1: Overview of selected clinical algorithms for ROP prediction.

Algorithm	Basic principle	Source
WINROP	Tracks weekly postnatal weight gain and raises an alarm when growth deviates from the expected trajectory.	Wu et al. [6]
G-ROP	Combines birth weight, gestational age and postnatal weight gain into a model with six criteria.	Binenbaum et al. [7]
ROPScore / CHOP ROP	Similar approaches based on clinical risk factors, validated on independent cohorts.	Piermarocchi et al. [8]

1.4 Motivation and Aim

The motivation for this topic came from the author's personal experience: her brother was born preterm and developed retinopathy of prematurity (ROP), which led to permanent vision loss. This experience was the direct reason for choosing the topic, while the methodology itself relies only on publicly available data and standard, verifiable machine learning procedures.

Unlike previous work on the same dataset [5], which selected genes with linear regularized methods, this thesis uses a combinatorial search based on a genetic algorithm and evaluates it with nested cross-validation, which fully isolates gene selection from the test data.

The aim of this thesis is to develop and evaluate a genetic algorithm-based method for selecting placental gene biomarkers associated with ROP risk, using the SHAP method to explain the model's predictions. The specific objectives are:

1. Carry out an independent differential gene expression analysis on a publicly available placental transcriptomic dataset to identify candidate genes.
2. Develop a genetic algorithm that selects an optimal gene subset for ROP risk classification, with nested cross-validation of performance to avoid the well-known problem of *selection bias* in small samples [11].
3. Compare the performance of the resulting model with standard feature selection approaches.
4. Apply the SHAP method to explain the contribution of individual genes to the model's predictions.

The thesis is engineering- and methodology-oriented: its contribution lies in applying and evaluating machine learning algorithms on structured biological data, while the medical problem defines the goal of the analysis and how the results are interpreted.

2

Literature Review

2.1 Placental Genes as ROP Biomarkers

In a cohort of preterm infants, Owen et al. [1] showed a significant association between acute placental inflammation and a reduced risk of ROP. This protective effect was strongest in infants born to mothers without preeclampsia. The exact biological mechanism behind this protective effect is not fully understood, but the authors suggest that mild placental inflammation may "prime" the fetal immune and vascular systems for postnatal conditions, making them more resistant to the later pathological processes that lead to ROP. Building on this finding, Huang et al. [5] analyzed the same placental transcriptomic dataset and identified four genes (*MMP25*, *CAPG*, *MCAM* and *B2M*) whose expression is statistically associated with ROP development. The authors selected these genes with a combination of differential expression analysis and elastic-net regularized regression. They then combined them with clinical variables into a nomogram, a graphical risk assessment tool that merges several factors into a single score. Although Huang et al. [5] showed that placental expression of these genes has predictive value, their gene selection approach relies on linear statistical methods. This thesis examines whether a nonlinear, combinatorial search based on a genetic algorithm can identify the same or a different, potentially more robust gene set, with more rigorous validation of performance.

2.2 Genetic Algorithms for Feature Selection in Genomics

Selecting a small number of relevant genes from high-dimensional datasets (typically thousands to tens of thousands of genes, with only tens to hundreds of samples) is one of the central challenges in bioinformatics. In this context, genetic algorithms are used as a *wrapper* feature selection method. Instead of scoring each gene individually, they evaluate the performance of entire *combinations* of genes and improve these combinations iteratively by simulating natural selection [12]. According to the review by Taha et al. [13], genetic algorithms are among the most widely used approaches in this field and can reach high classification accuracy with a relatively small number of selected genes. Deutsch [14] was among the first to apply evolutionary algorithms to find optimal gene sets for prediction from microarray data. He showed that this approach can reach classification accuracy comparable to or better than classical statistical methods, with far fewer selected genes. Similar approaches, including

hybrid methods that combine a genetic algorithm with support vector machines (GA-SVM) or k-nearest neighbors (GA-KNN), have been successful in classifying different cancer types from microarray data [12].

However, applying such methods to datasets with few samples (typically under 100) carries a significant methodological risk. Ambroise and McLachlan [11] showed that when gene selection and model evaluation are done on the same data, even with cross-validation, model accuracy is systematically overestimated. This is known as *selection bias*. The problem is especially pronounced for genetic algorithms. Through a large number of iterative evaluations, they effectively "search" the solution space for the solution that best fits a particular train/test split. This can lead to a gene combination that generalizes well within that specific split by chance, but not on truly new data. Given this risk, the genetic algorithm in this thesis was evaluated with nested cross-validation, where gene selection is fully isolated from the data used for the final performance estimate. This approach is described in detail in Chapter 3.

2.3 Explainable AI in Gene Expression Analysis

The SHAP (*SHapley Additive exPlanations*) method [15] is one of the most widely used approaches for explaining the predictions of machine learning models. It is based on the Shapley value from cooperative game theory. For each individual prediction, SHAP computes the contribution of every input feature, which allows both global interpretation (which features matter most on average) and local interpretation (why the model made a particular decision for a specific case). Applying SHAP to placental transcriptomic data has proven especially useful for other pregnancy-related conditions. Using a multi-algorithm machine learning approach on two placental transcriptomic datasets, Chen et al. [16] identified six genes linked to dysregulated ribosome biogenesis as biomarkers of preeclampsia risk. Their SHAP analysis showed that one gene (*SUV39H1*) is a dominant risk factor, while another (*GLUL*) has a protective effect. This approach, machine learning combined with SHAP explainability on small, publicly available placental transcriptomic datasets, is directly analogous to the method used in this thesis, except that the condition studied here is ROP rather than preeclampsia.

3

Data and Methodology

Figure 3.1 shows the overall workflow described in this chapter, from the raw data through preprocessing and feature selection to the final model and its evaluation through several complementary analyses.

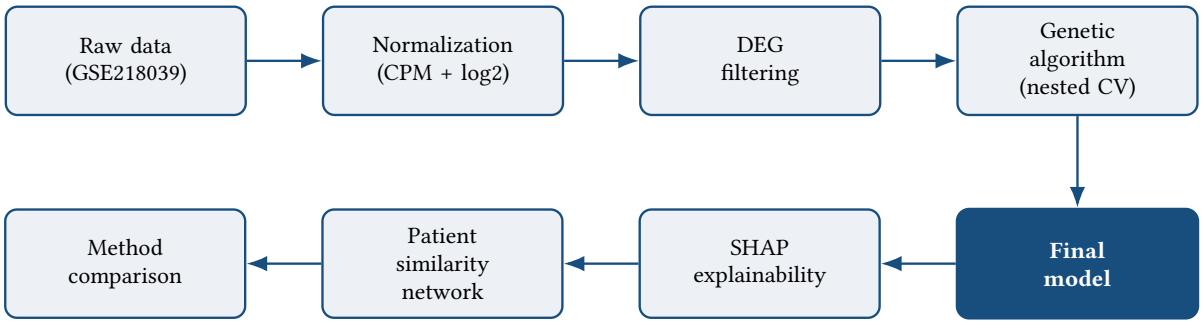


Figure 3.1: Overall workflow of the methodology used in this thesis.

3.1 Dataset

This thesis uses the publicly available dataset **GSE218039**, deposited in the Gene Expression Omnibus (GEO) database [1]. It contains RNA sequencing results for placental tissue from 30 preterm infants: 14 were diagnosed with ROP and 16 did not develop the disease. For each sample, expression data are available for 60,664 genes (given as the number of sequenced reads, i.e. *read counts*), together with accompanying clinical variables: the infant's sex and the presence of preeclampsia in the mother.

3.2 Preprocessing and Normalization

Raw gene expression data (*read counts*) are not directly comparable between samples, because the total number of reads per sample varies due to technical factors of the sequencing process rather than real biological differences. To remove this effect, the data were normalized with CPM (*counts per million*): the expression of each gene was divided by the total number of reads in that sample and multiplied by one million:

$$\mathrm{CPM}_{ij} = \frac{x_{ij}}{\sum_k x_{kj}} \times 10^6 \tag{3.1}$$

where x_{ij} is the number of reads of gene i in sample j. After normalization, a logarithmic transformation ($\log_2(\text{CPM} + 1)$) was applied. This is standard practice in gene expression analysis: it stabilizes the variance and compresses the wide range of expression values onto a scale better suited to further statistical analysis and machine learning.

Figure 3.2 illustrates the effect of normalization: the total number of reads per sample ranges from 11.4 to 25.8 million, while after CPM and $\log_2$ transformation the expression distributions are comparable across samples.

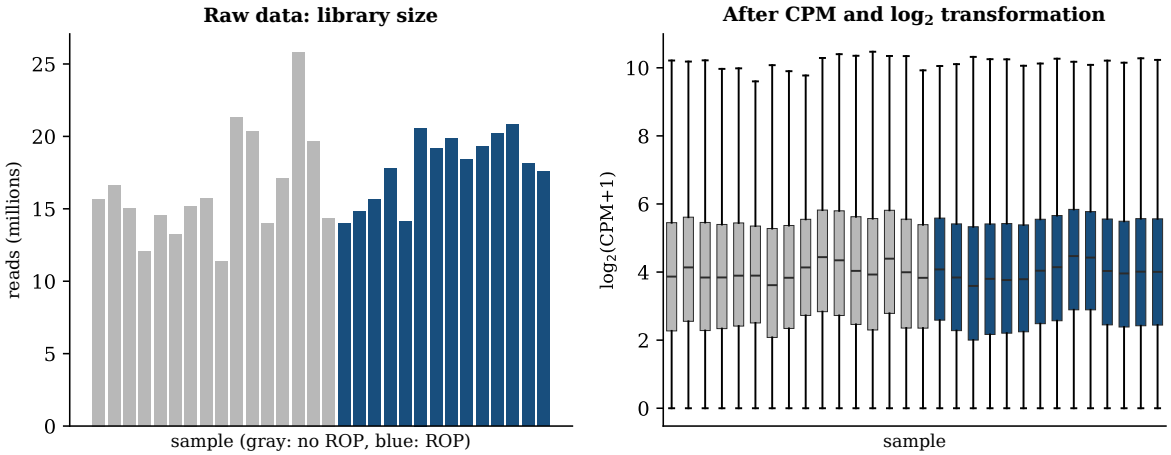


Figure 3.2: Left: total number of reads per sample in the raw data. Right: distribution of $\log_2$(CPM+1) values per sample for 15,307 expressed genes (CPM ≥ 1 in at least 15 samples). Gray: no ROP, blue: ROP.

3.3 Differential Expression Analysis

To reduce the search space for the genetic algorithm to a reasonable, biologically relevant size, a first filtering step was carried out as a differential expression analysis (DEG). For each of the 60,664 genes, Student's t-test was used to compare mean log-CPM expression between infants with and without ROP. After excluding genes with insufficient expression variability (for which the test could not be computed), the analysis covered 42,638 genes.

Because of the large number of statistical tests run at once, the Benjamini-Hochberg correction for multiple testing (*false discovery rate*, FDR) was applied, which controls the expected proportion of false positives. The final candidate set for further analysis was formed with a looser, uncorrected significance threshold ($p < 0.05$) combined with an effect size threshold ($|\log_2 FC| > 1$, where FC stands for *fold change*), because the strict corrected threshold ($p_{adj} < 0.05$) keeps only a handful of genes given the small sample size. This two-stage approach, filtering at a looser statistical threshold followed by more rigorous validation in later steps, is common practice in the analysis of genomic data with few samples.

These criteria selected 121 genes. The four genes identified in previous research [5] (*MMP25*, *CAPG*, *MCAM*, *B2M*) were added to this set regardless of whether they met the filtering criteria

on their own. Since *CAPG* was already among the 121 genes, the remaining three were added. This produced the final set of 124 candidate genes, which is the input search space for the genetic algorithm described in the next section (Section 3.4). Methodological details on re-running this procedure as part of performance estimation are given in Section 3.5.

3.4 Genetic Algorithm for Gene Selection

3.4.1 Solution Representation

Each candidate solution to the gene selection problem is represented by a binary vector (chromosome), where each position corresponds to one gene from the candidate set formed in the previous step. A value of 1 at position i means the corresponding gene is included in the combination, and 0 means it is excluded. In the illustrative run on all samples, the candidate set has 124 genes, so the chromosome length is 124:

$$c = (c_1, c_2, \ldots, c_{124}), \quad c_i \in \{0, 1\} \tag{3.2}$$

The chromosome length always equals the size of the candidate set given to that call of the genetic algorithm. Within nested validation (Section 3.5), it varies by outer iteration according to that iteration's DEG filtering (93–227 genes).

When the population is initialized, the number of included genes per chromosome is drawn at random from the range $[5, 20]$. This restricts the search space to biologically and clinically interpretable feature set sizes, avoiding both too few genes (not informative enough) and too many (hard to interpret).

3.4.2 Fitness Function

The quality of each chromosome c is measured by a *fitness function*, which maps a gene combination to a scalar value reflecting its predictive usefulness:

$$f(c) = \mathrm{AUC}(c) - \lambda \cdot |c| \tag{3.3}$$

where $\mathrm{AUC}(c)$ is the area under the ROC curve (*Area Under the ROC Curve*) of a model trained only on the genes included in chromosome c, estimated with cross-validation, $|c|$ is the number of included genes (the Hamming weight of the chromosome), and $\lambda = 0.001$ is a coefficient that mildly penalizes chromosomes with more selected genes. This formulation favors not only predictive accuracy but also parsimony (simplicity) of the selected feature set, following the principle that fewer genes make biological interpretation and clinical application easier. AUC can be read as the probability that the model assigns a higher risk to a randomly chosen infant with ROP than to a randomly chosen infant without ROP: a

value of 0.5 corresponds to random guessing, and 1 to perfect separation. To estimate AUC(c) during evolution, a Random Forest classifier with cross-validation is used (described in detail in Section 3.5), applied only to the training subset of the data. This prevents direct leakage of information from the test set into the feature selection process.

3.4.3 Genetic Operators

The population of chromosomes evolves across generations through three standard genetic operators.

Selection. Parent chromosomes are chosen by *tournament selection* with tournament size $k = 3$: k chromosomes are drawn at random from the population, and the one with the highest fitness among them wins and becomes a parent. Unlike directly picking the best chromosomes, this keeps the population diverse and lowers the risk of premature convergence to a local optimum.

Crossover. For a pair of selected parents $c^{(1)}$ and $c^{(2)}$, *uniform crossover* is applied: for each position $i \in \{1, \ldots, 124\}$, the offspring value $c_i^{(o)}$ is taken from one of the two parents at random with equal probability:

$$c_i^{(o)} = \begin{cases} c_i^{(1)} & \text{with probability } 0.5 \\ c_i^{(2)} & \text{with probability } 0.5 \end{cases} \tag{3.4}$$

Mutation. After crossover, mutation is applied to the offspring: for each position i, the value $c_i^{(o)}$ is flipped (from 0 to 1 or vice versa) with probability $p_m = 0.02$, independently for each position. Mutation introduces new genetic variation that cannot arise from crossover of existing chromosomes alone, which keeps the search from stagnating in a limited part of the solution space.

Figure 3.3 shows an example of selection, crossover and mutation on a shortened chromosome of 12 genes.

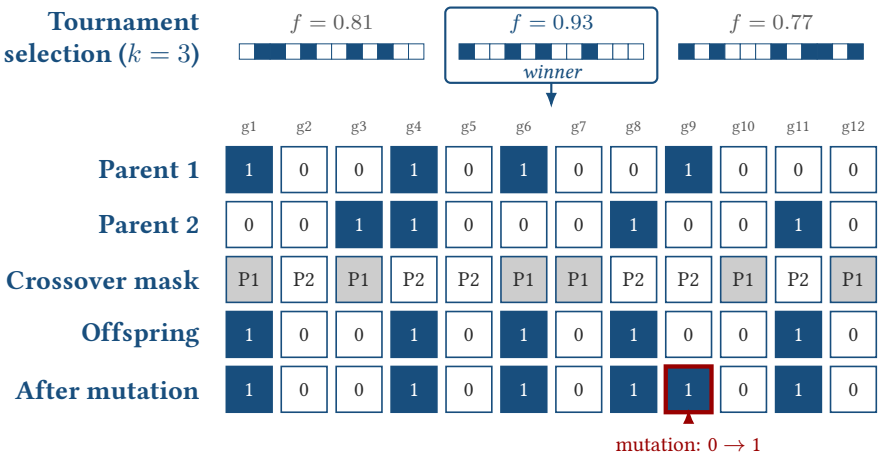


Figure 3.3: Example of genetic operators on a chromosome of 12 genes (124 in this thesis). Tournament selection picks the chromosome with the highest fitness among three randomly chosen ones ($k = 3$; values are illustrative). A blue cell marks an included gene. The crossover mask determines, for each position, whether the offspring takes the gene from the first (P1) or the second parent (P2), and mutation then flips one position (red outline).

Elitism. To preserve the best solution found, the chromosome with the highest fitness in each generation is carried over unchanged to the next generation, independently of selection, crossover and mutation.

3.4.4 The Complete Algorithm

Algorithm 1 summarizes the complete genetic algorithm used in this thesis. A population of size $N = 30$ evolves over $G = 30$ generations. In each generation, besides the elite chromosome, the remaining $N - 1$ offspring are produced with the selection, crossover and mutation described in the previous section.

Algorithm 1 Genetic algorithm for gene selection

Initialize a population P of N random chromosomes
for generation $g = 1$ to G **do**
    for each chromosome $c \in P$ **do**
        Compute $f(c)$ (Equation 3.3)
    $c^* \leftarrow$ chromosome with the highest $f(c)$ in P
    $P' \leftarrow \{c^*\}$ ▷ elitism
    while $|P'| < N$ **do**
        $c^{(1)} \leftarrow \text{TournamentSelection}(P)$
        $c^{(2)} \leftarrow \text{TournamentSelection}(P)$
        $c^{(o)} \leftarrow \text{Crossover}(c^{(1)}, c^{(2)})$
        $c^{(o)} \leftarrow \text{Mutation}(c^{(o)})$
        $P' \leftarrow P' \cup \{c^{(o)}\}$
    $P \leftarrow P'$
return the chromosome with the highest $f(c)$ in the final population P

Convergence of the algorithm is illustrated in Figure 4.4 (Chapter 4), which tracks the fitness of the best chromosome in the population across generations for several combinations of hyperparameters (N, p_m), analyzed as part of the sensitivity study (Section 4.3).

3.5 Nested Cross-Validation

Estimating the predictive performance of a model that includes a feature selection step needs special methodological care. If feature selection and performance estimation are done on the same data, even with standard k-fold cross-validation, the result is *selection bias* [11], which systematically overestimates the model's true predictive ability. This risk is especially pronounced for genetic algorithms. Through a large number of iterative evaluations (up to 900 per run in this thesis), they effectively search the solution space for the solution that best fits a particular data split. The result can be a feature combination that generalizes well only within that specific split.

To avoid this risk, **nested cross-validation** was used. It sets up two independent levels of data splitting:

- **Outer loop** (5-fold, stratified): the 30 samples are split into 5 subsets. In each iteration, one subset (6 samples) is held out only for the final performance estimate, while the remaining 4 subsets (24 samples) form that iteration's training set.
- **Inner loop** (4-fold): within the training set of each outer iteration, the genetic algorithm runs a full evolution (including re-running the DEG analysis from Section 3.3 on those 24 samples only), and the fitness of each chromosome is estimated with inner cross-validation.

Figure 3.4 shows this procedure.

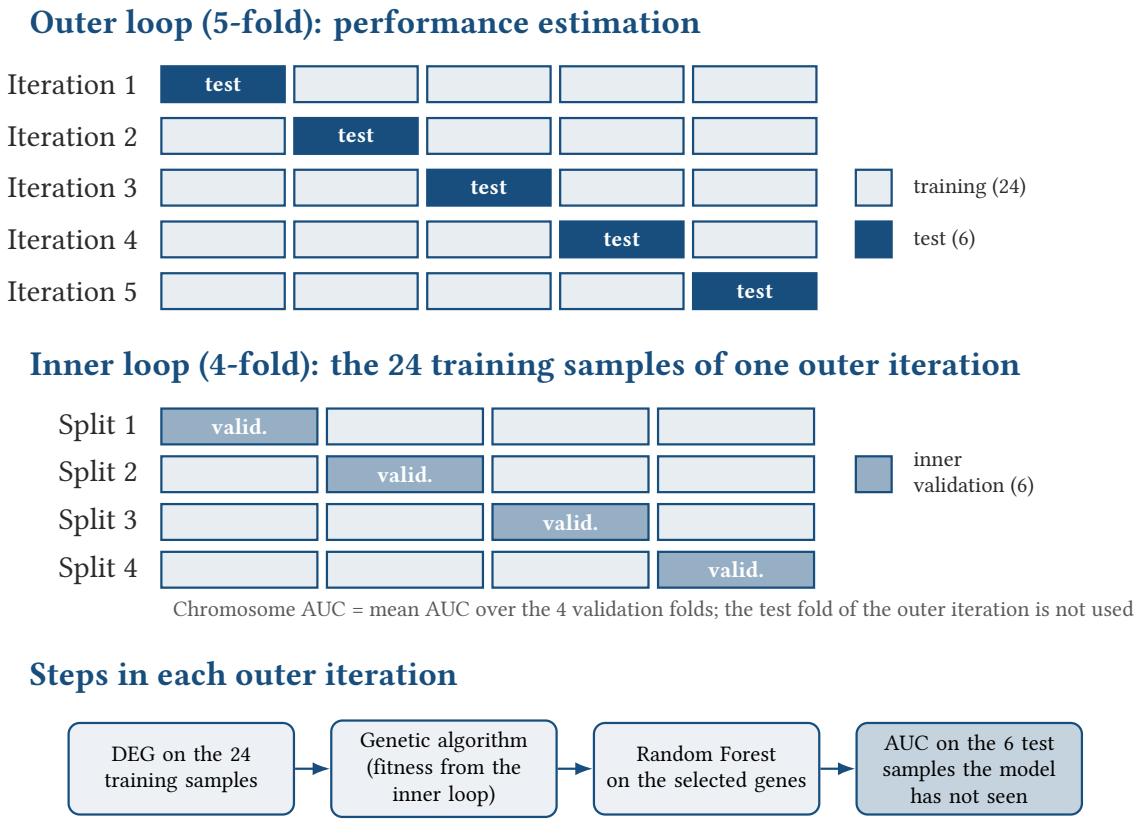


Figure 3.4: Nested cross-validation scheme. The outer loop splits the 30 samples into five folds and holds out 6 samples for final testing in each iteration. The inner loop splits the remaining 24 training samples into four folds and is used only to estimate fitness inside the genetic algorithm. The test samples of an outer iteration take part neither in gene selection nor in training.

After evolution finishes within an outer iteration, the chromosome with the highest fitness is used to train that iteration's final model. Its performance is then measured on the held-out test subset (6 samples), data that neither the genetic algorithm nor the training process used at any point. This procedure is repeated for all 5 outer iterations, and the reported performance is the mean AUC over the 5 independent test subsets.

Note that the size of the candidate gene set produced by DEG filtering varies between outer iterations (from 93 to 227 genes in the experiments run), because the DEG analysis is performed on different training subsets of 24 samples each. This variability is an expected consequence of re-running feature selection on different subsets of a small dataset.

3.6 Comparison with Alternative Feature Selection Methods

To assess the contribution of the genetic algorithm objectively, its performance was compared with four standard feature selection methods applied under identical conditions: the same

nested validation protocol described in the previous section, including re-running the DEG analysis within each outer iteration:

- **Random Forest (baseline)**: a classifier trained on the full candidate gene set, without an extra feature selection step, used as a reference point.
- **LASSO** (L_1 regularization): logistic regression with an L_1 penalty on the coefficients, which performs implicit feature selection by shrinking the coefficients of unimportant genes to exactly zero.
- **Ridge** (L_2 regularization): logistic regression with an L_2 penalty, used as an additional reference point without explicit feature selection.
- **Elastic-net** (a combination of L_1 and L_2 regularization): the method used in previous research [5] to select placental ROP biomarkers.

For the regularization-based methods, the regularization strength (parameter C) and, for elastic-net, the ratio of the L_1 and L_2 components (parameter $l1_ratio$) were chosen by *grid search* within the inner loop of nested validation, analogous to the optimization of the genetic algorithm.

3.7 SHAP Explainability

To interpret the predictions of the final model, the SHAP (*SHapley Additive exPlanations*) method [15] was used, based on Shapley values from cooperative game theory. For each individual prediction, SHAP assigns each input feature (gene) a contribution value that represents its effect on how far the prediction deviates from the model's expected (average) value:

$$\hat{y}(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{3.5}$$

where $\hat{y}(x)$ is the model's prediction for sample x, ϕ_0 is the expected (average) prediction of the model over the whole dataset, ϕ_i is the SHAP value of gene i for that sample, and M is the total number of genes in the model. SHAP values were computed with `TreeExplainer`, an implementation optimized for tree-based models (including Random Forest), which computes exact (not approximate) Shapley values for this class of models.

The final model used for SHAP analysis was trained by running the genetic algorithm on the full set of 30 samples (Section 3.4). It is important to clearly separate two different uses of models in this thesis. The *evaluation model* is trained only on training subsets within nested validation (Section 3.5), and its performance on unseen test subsets is the reported accuracy (Section 4.2). The *final interpretation model* is trained on the full dataset after evaluation has already been completed independently, and its only purpose is SHAP analysis, not an additional accuracy estimate.

3.8 Patient Similarity Network

As an additional, independent check for signal in the selected gene set, a patient similarity network (*Patient Similarity Network*, PSN) [17] was built. Its nodes represent the 30 samples (infants), and its edges connect samples with similar expression profiles of the final gene set.

The similarity between two samples x_j and x_k was computed with a Gaussian kernel over the Euclidean distance between their standardized expression vectors:

$$s_{jk} = \exp\left(-\frac{\|x_j - x_k\|^2}{2\sigma^2}\right) \tag{3.6}$$

where σ is the median of all pairwise distances in the dataset. From this similarity matrix, a k-nearest-neighbor graph ($k = 5$) was built, in which each sample is connected to the 5 samples most similar to it. Louvain community detection was applied to this graph, and the agreement between the detected communities and the actual ROP status was quantified with the Adjusted Rand Index (ARI).

3.9 Implementation

All analyses in this thesis were implemented in Python 3.11, in a Jupyter environment, as a series of modules that follow the workflow in Figure 3.1. For clarity, Figure 3.5 shows the same procedure from the software organization point of view: four layers, from data loading to interpretation, where each layer uses the output of the previous one.

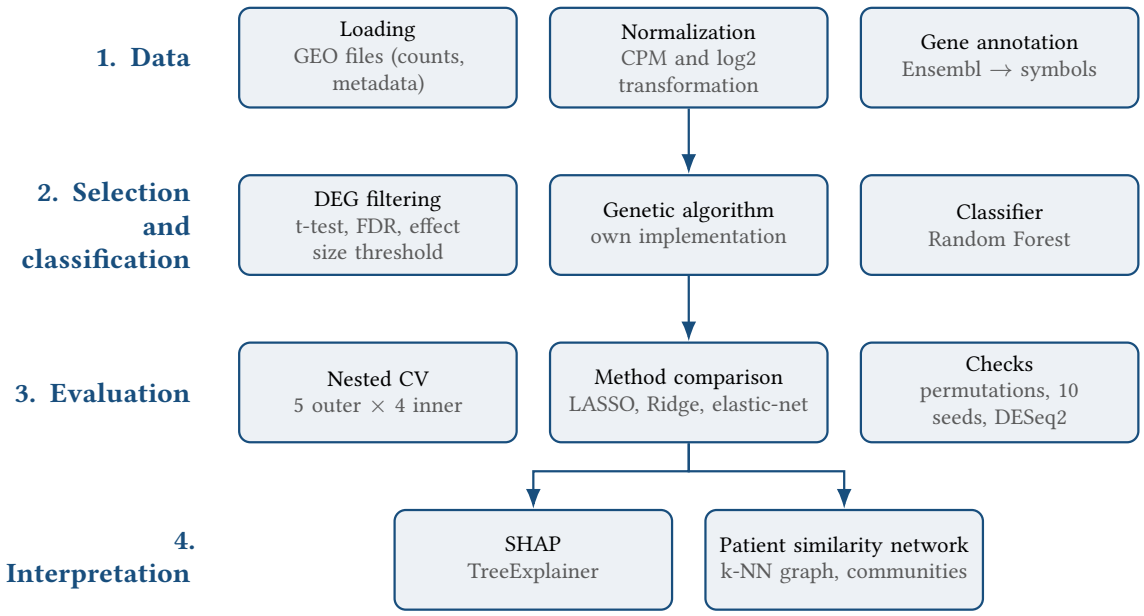


Figure 3.5: Layered organization of the implementation. Each layer corresponds to one phase of the analysis and uses the output of the previous layer.

Table 3.1 lists the libraries used in each layer. The genetic algorithm is the only part of the

procedure implemented from scratch: selection, crossover, mutation, elitism and the fitness function are written as separate functions over binary vectors, using the standard Python *random* module and *NumPy*, with no specialized evolutionary algorithm library. The fitness function calls the classifier and cross-validation from the *scikit-learn* library.

Table 3.1: Libraries used in the implementation and their role.

Library	**Role**
pandas, NumPy	loading expression tables, CPM normalization and log2 transformation, feature matrices
SciPy, statsmodels	Student's t-test and Mann-Whitney test, Benjamini-Hochberg correction
scikit-learn	Random Forest, logistic regression (L1, L2, elastic-net), grid search, stratified cross-validation, AUC
SHAP	*TreeExplainer* for computing Shapley values
mygene	annotating Ensembl identifiers with gene symbols
pydeseq2	exploratory check with the DESeq2 method (Section 5.3)

The most computationally demanding part of the procedure is the fitness function. One run of the genetic algorithm involves 900 chromosome evaluations (30 chromosomes over 30 generations), and each evaluation trains a Random Forest with 100 trees inside 4-fold cross-validation. Within nested validation, shown in Algorithm 2, the algorithm runs once for each of the five outer iterations, which gives $5 \times 900 \times 4 = 18\,000$ model fits for a single performance estimate. The stability analysis (10 runs with different seeds) and the permutation test (500 permutations) add tens of thousands more. For this reason, the fitness function used fewer trees (100) than the final model (200). All runs were performed on a standard personal computer without a GPU.

Algorithm 2 Nested cross-validation with gene selection

Split the 30 samples into 5 stratified subsets ▷ outer loop
for each subset $k = 1$ to 5 **do**
    $S_{test} \leftarrow$ subset k; $S_{train} \leftarrow$ remaining 4 subsets
    Run DEG analysis on S_{train} and form the candidate set K_k (including the four genes from the literature)
    $c_k^* \leftarrow \text{GeneticAlgorithm}(S_{train}, K_k)$ ▷ fitness: AUC, 4-fold CV
    Train a Random Forest on S_{train} using the genes in c_k^*
    $\text{AUC}_k \leftarrow$ AUC on S_{test}
return mean and standard deviation of $\text{AUC}_1, \ldots, \text{AUC}_5$

Reproducibility was ensured by fixing the random number generators: *random_state* was set to 42 for the Random Forest and for cross-validation splits, and the Python and NumPy generators were initialized with the same seed in the final run, and with seeds 0 to 9 in

the stability analysis. Fixed seeds guarantee identical results only in the same software environment: minor differences can appear between library versions (e.g. scikit-learn and SHAP), so exact reproduction requires keeping the versions used in this thesis.

4

Results

4.1 Differential Gene Expression

Table 4.1 shows the ten genes with the lowest p-values among those that meet the effect size threshold $|\log_2 FC| > 1$ (Section 3.3), together with their log2 fold change (log2FC) values. A positive log2FC means higher expression of the gene in infants with ROP compared with the control group.

Table 4.1: The ten genes with the lowest p-values among genes with $|\log_2 FC| > 1$.

Gene	log2FC	p-value	p_{adj}
HELLPAR	1.62	2×10^{-6}	0.041
AQP11	-1.00	3×10^{-6}	0.041
RTN4RL2	1.33	9×10^{-6}	0.095
HLA-V	1.02	1.1×10^{-5}	0.095
TLE6	1.02	1.4×10^{-5}	0.096
PLAC8	2.21	6.0×10^{-5}	0.243
NOG	2.17	6.3×10^{-5}	0.243
HLA-G	2.93	8.5×10^{-5}	0.252
ASAP3	1.25	1.3×10^{-4}	0.308
CLDN19	1.17	1.6×10^{-4}	0.321

Of the 42,638 genes tested, 3 meet the strict corrected significance threshold ($p_{adj} < 0.05$): *HELLPAR*, *AQP11* and *HCG4*. The last one is not shown in Table 4.1 because it does not meet the effect size threshold ($\log_2 FC = 0.69$). With the looser criterion described in Section 3.3 (uncorrected $p < 0.05$ and $|\log_2 FC| > 1$), the candidate set contains 121 genes. Of the four genes identified in previous research [5] (*MMP25*, *CAPG*, *MCAM*, *B2M*), only *CAPG* meets this criterion on its own. The other three are not among the 121 genes, but they were added to the candidate set based on the previous research on the same cohort, which gave the final set of 124 genes.

Figure 4.1 shows the results of the DEG analysis for all tested genes. Of the 121 candidates, 99 have higher and 22 lower expression in infants with ROP. The two genes with the lowest p-values, HELLPAR ($\log_2 FC = 1.62$) and AQP11 ($\log_2 FC = -1.00$), are the only ones that remain significant after correction for multiple testing ($p_{adj} = 0.041$). The same genes are also the ones the genetic algorithm selected most consistently (Section 4.4). The other genes in the

final set come from a wider range of the DEG ranking, which shows that besides the strongest individual signals, the algorithm also selects genes that contribute in combination with others.

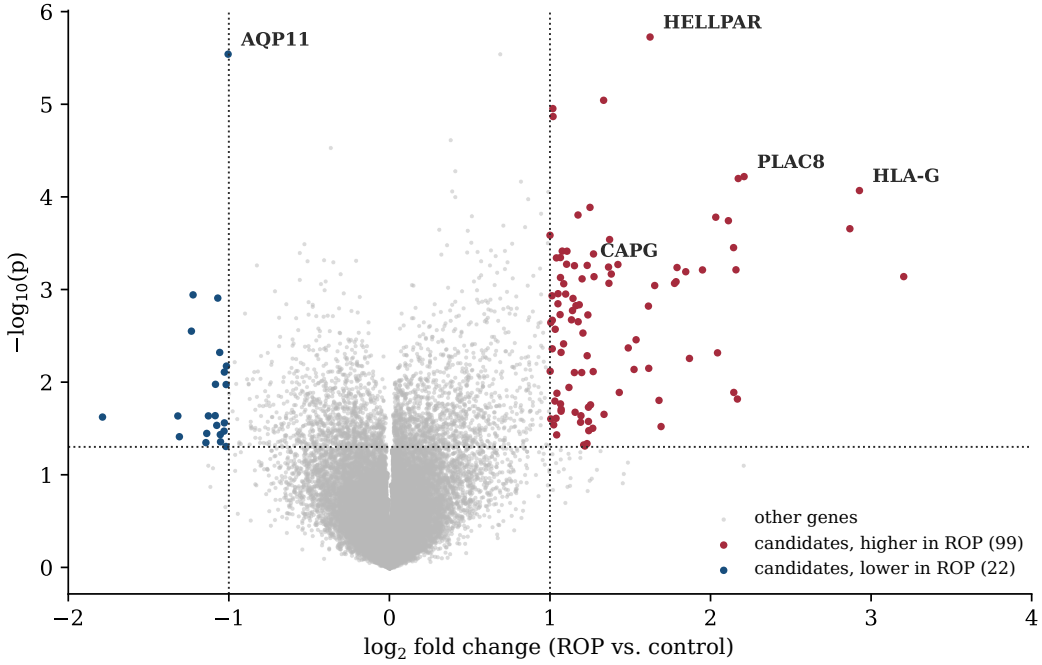


Figure 4.1: Volcano plot of the differential expression analysis (42,638 genes). Dotted lines mark the thresholds $p = 0.05$ and $|\log_2 FC| = 1$; colored points are the 121 candidate genes.

4.2 Genetic Algorithm Performance and Method Comparison

Table 4.2 summarizes the performance of the genetic algorithm and four alternative feature selection methods, estimated with the identical nested cross-validation protocol described in Section 3.5. Figure 4.2 shows the same results visually.

Table 4.2: Performance comparison of feature selection methods (AUC, nested CV, $n = 5$ outer iterations).

Method	AUC (mean ± std)	Number of features
Genetic algorithm	**0.919 ± 0.068**	**6–10 (mean 8.2)**
Random Forest (baseline)	0.878 ± 0.194	124 (fixed)
Ridge (L2)	0.811 ± 0.171	all candidates
Elastic-net	0.744 ± 0.191	43–135
LASSO (L1)	0.656 ± 0.286	9–16

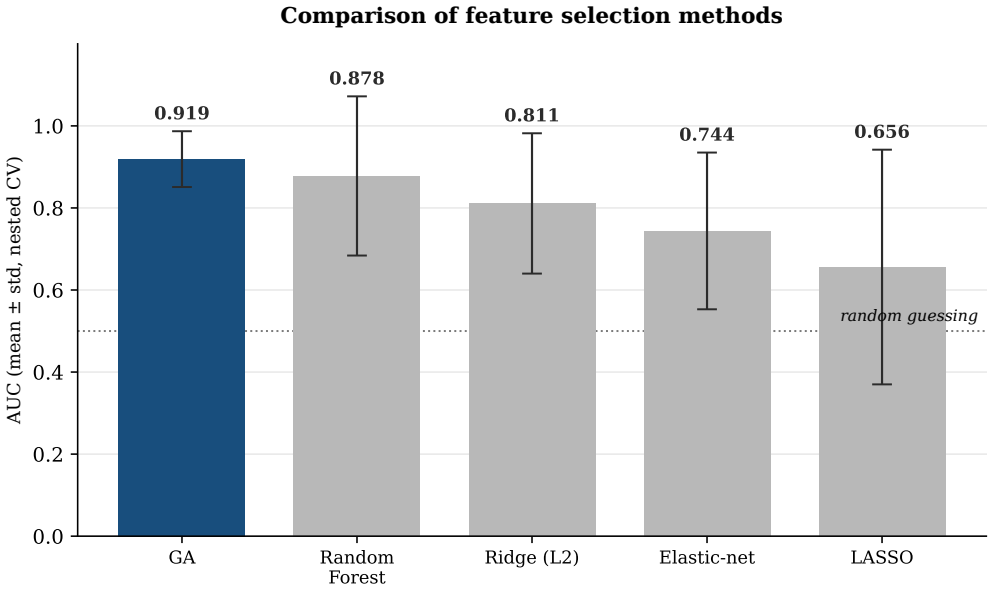


Figure 4.2: AUC comparison of the genetic algorithm and alternative feature selection methods. Bars show mean values and standard deviation over the 5 outer iterations of nested validation.

The genetic algorithm reaches the highest mean AUC and, at the same time, the lowest standard deviation of all tested methods. This points to more stable performance across different data splits, in addition to higher accuracy with far fewer selected genes than any alternative approach.

Table 4.3 shows the AUC of the genetic algorithm in each outer iteration, illustrating the performance variability typical of small test subsets (6 samples per iteration).

Table 4.3: AUC of the genetic algorithm per outer iteration of nested validation.

Outer iteration	AUC	Number of selected genes
1	0.875	10
2	0.833	6
3	1.000	7
4	0.889	9
5	1.000	9

4.2.1 Permutation Test

To formally check that the measured predictive ability is not due to chance, a permutation test was run: the ROP labels were randomly permuted 500 times, and for each permutation the AUC of the final gene set was computed with the same cross-validation. The mean AUC on permuted labels was 0.485 ± 0.152, close to the value of 0.5 that corresponds to random guessing, as expected when no real link between the features and the outcome exists. The reference AUC in this test (1.0) was computed on the same data on which the genes were selected, so it

is overestimated; the honest performance estimate is given in Section 4.2 (AUC $= 0.919$). The proportion of permutations that reached or exceeded this more conservative value was still $p = 0.002$, which confirms that the measured signal deviates significantly from the null distribution regardless of which reference value is used.

4.2.2 Effect of Previously Known Genes

The four genes from previous research [5] were identified on the same dataset (GSE218039), so adding them by hand to the candidate set of each outer iteration (Section 3.3) in theory introduces a slight link to that iteration's test data. To check the effect of this decision, the whole nested procedure was repeated without manually adding these four genes, relying only on the DEG results of each outer iteration. The mean AUC was 0.908 ± 0.084, almost identical to the original result (0.919 ± 0.068), which indicates that adding these four genes by hand had no practical effect on the measured performance.

Figure 4.3 shows the expression of the 10 genes of the final model (Section 3.7) across all 30 samples.

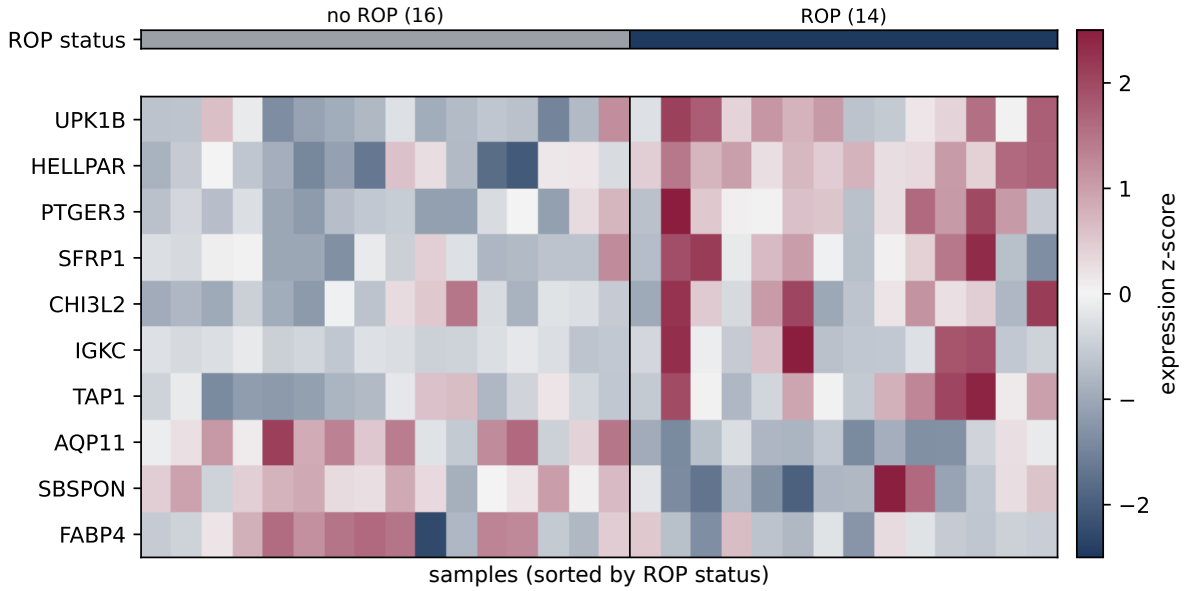


Figure 4.3: Expression of the 10 genes of the final model (z-score per gene), with samples sorted by ROP status. The genes were selected on all 30 samples, so the visual difference between groups is not an estimate of predictive performance; that estimate is given in Section 4.2.

4.3 Convergence and Hyperparameter Sensitivity

Figure 4.4 shows the convergence of the best chromosome's fitness across generations for three mutation probabilities ($p_m \in \{0.01, 0.05, 0.15\}$), with a fixed population size of $N = 30$.

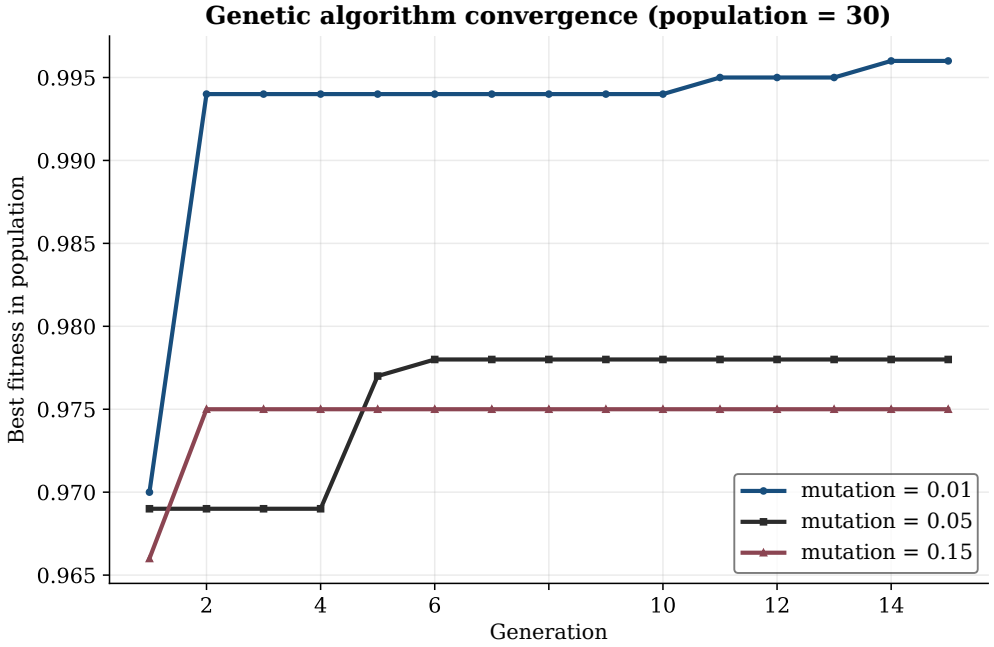


Figure 4.4: Convergence of the genetic algorithm for different mutation probabilities. A lower mutation rate ($p_m = 0.01$) gives more gradual but better long-term convergence than higher mutation rates, which stall early at a local optimum.

Table 4.4 summarizes the final fitness values for all tested combinations of population size and mutation probability, after 15 generations.

Table 4.4: Sensitivity of final fitness to the genetic algorithm's hyperparameters.

Population (N)	**Mutation (p_m)**	**Final fitness**
10	0.01	0.990
10	0.05	0.964
10	0.15	0.953
30	0.01	0.996
30	0.02	0.987
30	0.05	0.978
30	0.15	0.975
60	0.01	0.997
60	0.05	0.983
60	0.15	0.994

The results show that the lowest tested mutation rate ($p_m = 0.01$) gives the best final solution at every population size, while increasing the population from 30 to 60 brings a negligible improvement ($+0.001$) at twice the computational cost. The value $p_m = 0.02$ used in the main experiments of this thesis gives a final fitness of 0.987, between the best ($p_m = 0.01$, 0.996) and the middle ($p_m = 0.05$, 0.978) tested values, which confirms that it was a reasonable choice for the main experiments.

4.4 Gene Selection Stability

To assess the robustness of gene selection, the genetic algorithm was run independently 10 times with different random seeds on the full set of 30 samples. Table 4.5 shows the genes selected in at least 5 of the 10 runs.

Table 4.5: Genes with the highest selection rate across 10 independent runs of the genetic algorithm.

Gene	Number of runs (out of 10)
HELLPAR	10
AQP11	7

No other gene in the candidate set reached the threshold of 5 runs, including none of the four genes identified in previous research [5]. This instability of feature selection between individual runs is consistent with the known limitations of genomic analyses on small samples [11], while HELLPAR stood out in the main configuration as the most stable feature (selected in all 10 runs). This is discussed further in Chapter 5.

4.5 Exploratory Patient Similarity Analysis

Figure 4.5 shows the patient similarity network built from the expression of the final set of 10 genes (Section 3.8). The resulting graph is connected (30 nodes, 102 edges, a single connected component).

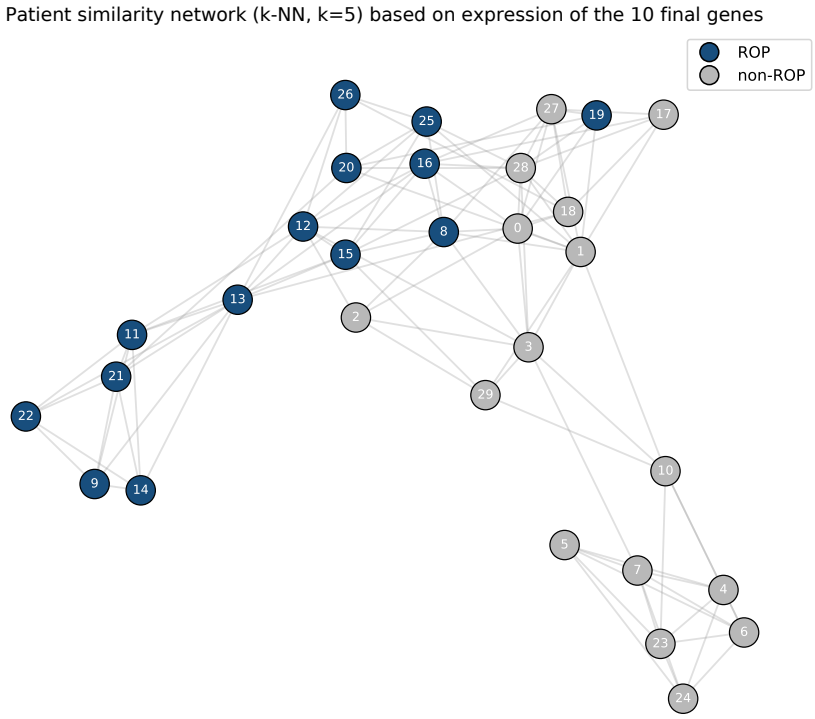


Figure 4.5: Patient similarity network based on the expression of the final gene set. Nodes represent samples, colored by ROP status (dark blue: ROP, gray: no ROP); edges connect samples with the most similar expression profiles ($k = 5$).

Louvain community detection identified 4 communities in the network. The agreement between these communities and the actual ROP status, measured by the Adjusted Rand Index, is ARI $= 0.173$. This is an exploratory, hypothesis-generating finding, not a formal model validation. Unsupervised clustering does not use the clinical labels, and the positive, moderate ARI suggests structure consistent with ROP status, while also reflecting other sources of biological variability in the data.

4.6 SHAP Explainability

Figure 4.6 shows the SHAP values of the final model (Section 3.7) for all 10 genes, sorted by decreasing mean absolute impact on the prediction.

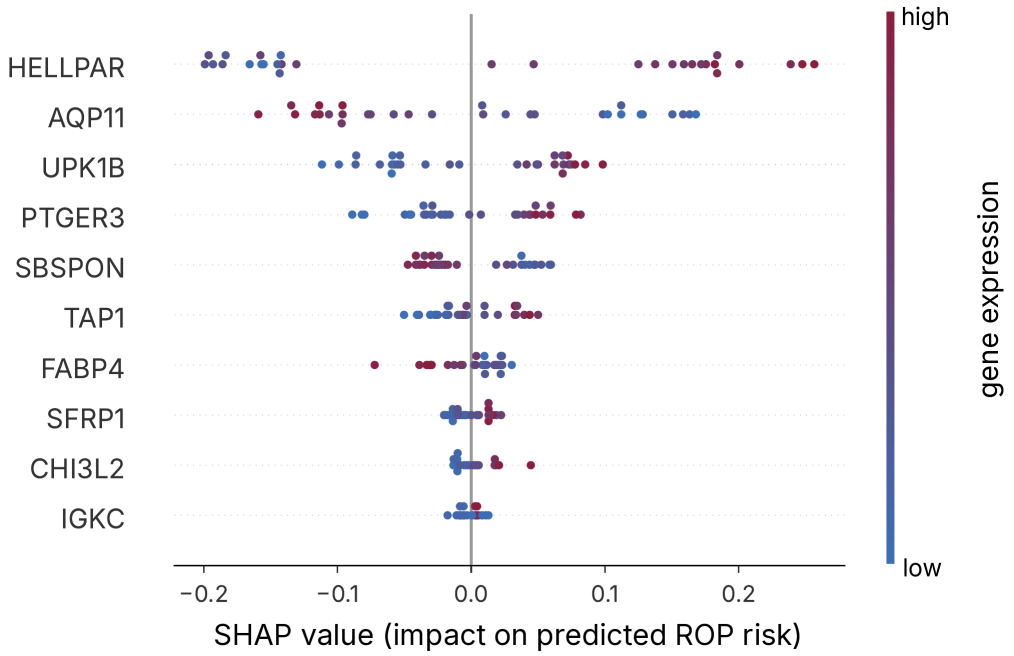


Figure 4.6: SHAP explainability of the final model. Each point is one sample; its position on the horizontal axis shows the impact on the predicted risk, and its color shows the expression value of that gene in the sample (blue: low, burgundy: high).

The gene HELLPAR shows the largest and most consistent impact on the model's predictions. Samples with high expression (burgundy points) almost exclusively push the predicted risk up, while samples with low expression (blue points) consistently push it down, with clear separation and only a few samples near zero. This consistency is absent for the other genes in the model. The combination of the highest selection stability across independent runs of the genetic algorithm in the main configuration (Section 4.4) and the strongest, most consistent contribution to the prediction in the SHAP analysis makes HELLPAR the most prominent finding of the main analysis. Its biological interpretation is discussed in Chapter 5. Possible confounding by clinical variables was checked. HELLPAR expression does not differ significantly between sexes ($p = 0.47$, Mann-Whitney test) or between samples with and without maternal preeclampsia ($p = 0.46$, Mann-Whitney test). Sex is not significantly imbalanced between the ROP and control groups ($p = 0.73$, Fisher's exact test), and neither is preeclampsia (6 of 15 versus 8 of 15 ROP cases, $p = 0.72$, Fisher's exact test).

5

Discussion

This thesis examined whether a genetic algorithm can identify a small, predictive set of placental genes associated with ROP risk, with a methodologically rigorous evaluation of performance. The main findings can be summarized as follows: (1) in nested cross-validation, the genetic algorithm reached a higher mean AUC (0.919 ± 0.068) with fewer features than all tested alternative methods; (2) the permutation test, excluding the manually added genes from the literature, and replacing the DEG method (at the same candidate set size) do not suggest that this result is due to chance or to these choices; (3) in the main configuration of the analysis, the gene HELLPAR stood out as the most stable and most influential feature across independent runs of the algorithm and in the SHAP analysis.

5.1 Comparison with Previous Research

Unlike Huang et al. [5], who used elastic-net regularized regression for gene selection, this thesis applies a genetic algorithm as a combinatorial, nonlinear alternative. A direct comparison under identical nested validation conditions (Section 4.2) shows that the genetic algorithm outperforms elastic-net on this dataset (0.919 versus 0.744), with far fewer selected genes. Note that the four genes identified by Huang et al. [5] (*MMP25*, *CAPG*, *MCAM*, *B2M*) were previously derived from the same dataset (GSE218039) used in this thesis. Including them in the candidate set is therefore not independent external confirmation, but the use of prior knowledge from the same cohort. The check of this decision (Section 4.2.2) shows that excluding these genes from the candidate set changes performance only negligibly (0.908 versus 0.919). This indicates that the main result of this thesis does not depend mainly on that prior knowledge, and that the genetic algorithm finds an informative feature set directly from the data on its own.

None of the four genes identified by Huang et al. [5] was kept in the final gene set of this thesis (Section 4.4), and none appears among the genes selected consistently across independent runs of the algorithm. This is consistent with the well-documented instability of gene signatures derived from small samples: different, methodologically sound analyses of the same dataset can identify different but equally predictive feature sets, because genes within the same biological pathways are highly correlated.

5.2 Biological Interpretation: the HELLPAR Gene

HELLPAR is the only gene that was selected consistently in all 10 independent runs of the genetic algorithm in the main configuration of the analysis (Section 4.4), and it also shows the strongest and most consistent contribution to the prediction in the SHAP analysis (Section 4.6). It also has the strongest individual difference in expression between the groups in the DEG analysis. Three criteria, statistical difference, selection stability and impact on the prediction, point to the same gene. These criteria are not independent of each other, because they were computed on the same samples and the genetic algorithm selects from a set already filtered by statistical significance. Their agreement therefore shows that the analysis is consistent, not that the finding is independently confirmed.

HELLPAR is a long non-coding RNA (lncRNA) located in the region of chromosome 12q23 that is linked to HELLP syndrome, a severe form of preeclampsia [18]. It is expressed in the extravillous trophoblast, the placental cells that invade the uterine wall, and is associated with the regulation of their cell cycle and invasion [18]. Higher HELLPAR expression in infants who developed ROP may therefore reflect altered placental function. This fits the hypothesis that placental factors affect ROP risk (Section 1.2).

The second most stable gene, AQP11 (aquaporin 11), has lower expression in infants with ROP. This channel in the endoplasmic reticulum transports water and hydrogen peroxide and helps maintain the cell's redox balance [19], which links it to oxidative stress, a mechanism known from the first phase of ROP pathogenesis (Section 1.1).

It is important to state the scope of these findings precisely. In this dataset, HELLPAR expression does not differ between samples with and without preeclampsia (Section 4.6), so the signal is not simply a proxy for maternal preeclampsia. Biological plausibility is not evidence of causation. The finding should be interpreted as the basis for a hypothesis about the role of HELLPAR in ROP risk, which requires independent validation, and not as a confirmed biomarker.

5.3 Limitations

The results of this thesis are subject to several important limitations that must be considered when interpreting them.

Sample size. The dataset contains only 30 samples. This is typical of publicly available transcriptomic data of this kind, but it is a fundamental limitation for drawing general conclusions. The test subsets in nested validation contain only 6 samples, so the individual per-iteration AUC values (Table 4.3) are coarse, discrete estimates with limited statistical power.

Gene selection stability. As shown in Section 4.4, gene selection between independent runs of the genetic algorithm is unstable: only two genes (*HELLPAR*, *AQP11*) reach the threshold

of 5 out of 10 runs. The final set of 10 genes, used for the SHAP analysis and the patient similarity network, is the result of one specific run of the algorithm (fixed by a random seed), not a consensus, robust finding. This limitation is inherent to feature selection on small, high-dimensional datasets [11]. It does not undermine the validity of the measured performance (which was estimated independently with nested validation), but it calls for caution when interpreting the identity of individual selected genes.

Permutation test. The permutation test (Section 4.2) assesses the significance of the final, fixed gene set, but it does not repeat the full feature selection procedure (DEG and the genetic algorithm) for each permutation, because of the computational cost of such an approach. The test result should therefore be read as confirmation that the selected gene set carries signal above chance level, not as formal proof of the significance of the entire feature selection procedure.

No external validation. All results refer to a single, publicly available dataset. Independent validation on an external cohort is necessary before any conclusion about the clinical applicability of the identified features.

Methodological choices open to further study. The differential expression analysis in this thesis uses Student's t-test on log-transformed CPM values. Specialized RNA-seq methods (e.g. DESeq2, limma-voom), which explicitly model the mean-variance relationship typical of count data, could be a more refined choice. Similarly, the inner cross-validation inside the genetic algorithm's fitness function uses a fixed fold split, which in theory allows the algorithm to adapt slightly to that specific split over a large number of iterative evaluations. Finally, the hyperparameters of the genetic algorithm (Section 4.3) were tested individually, without repetition across multiple random seeds, so they represent an engineering sensitivity analysis rather than formal optimization.

A preliminary, exploratory check replaced the t-test with the DESeq2 method in the DEG filtering step. When the candidate set was limited to the same size as in the main analysis (the 121 genes with the lowest p-values with $|\log_2 FC| > 1$, plus the four known genes), the nested AUC was 0.883 ± 0.079, within the range of the main result (0.919 ± 0.068). With the strict threshold $p_{adj} < 0.05$, the candidate sets were larger (183–473 genes per iteration), and AUC was lower and more variable (0.789 ± 0.218). The identity of the most frequently selected genes differed from the main analysis. The performance estimate is therefore not strongly sensitive to the choice of DEG method at the same candidate set size, but the identity of the consistently selected genes is sensitive to the configuration of the DEG step, so HELLPAR should be interpreted as a hypothesis that holds for the main configuration.

5.4 Future Work

Based on the identified limitations, future work should include: validation of the identified features on an independent, external cohort; repeating nested validation with several random

data splits (*repeated nested cross-validation*) for a more precise estimate of performance variability; applying specialized differential expression methods for RNA-seq; and an explicit analysis of the contribution of the available clinical variables (sex, preeclampsia) relative to the gene signal, to check whether placental gene expression carries predictive value independently of known clinical risk factors.

6

Conclusion

This thesis developed and evaluated a genetic algorithm-based method for selecting placental gene biomarkers associated with the risk of retinopathy of prematurity, using the SHAP method to explain the predictions. An independent differential expression analysis was carried out on a publicly available placental transcriptomic dataset. Based on it, the genetic algorithm identified a small, predictive gene set, whose performance was estimated with methodologically rigorous nested cross-validation to avoid the well-known problem of *selection bias* typical of analyses with few samples.

In an honest evaluation, the genetic algorithm reached a mean AUC of 0.919, outperforming all tested alternative feature selection methods (Random Forest, LASSO, Ridge, elastic-net) with far fewer selected genes. The permutation test, excluding the manually added genes from the literature, and replacing the DEG method at the same candidate set size do not suggest that this result is due to chance or to these choices. In the main configuration of the analysis, the gene HELLPAR stood out as the only consistently stable and most influential finding across independent runs of the algorithm and in the SHAP analysis. It is a placental non-coding RNA linked to HELLP syndrome, which gives a biologically plausible link between placental function and ROP risk. The finding still requires independent validation before any conclusion about its biological role.

The thesis is primarily a methodological contribution. It shows that a combinatorial, evolutionary search of the feature space can identify a compact, predictive gene set on a small, high-dimensional transcriptomic dataset, with rigorous control of the known sources of performance overestimation. The limitations of the thesis, above all the small number of samples and the instability of individual gene selection between independent runs, clearly restrict the scope of the conclusions to a methodological *proof of concept*, not clinical validation.

Future work should include validation on an independent external cohort, repeated nested validation for a more precise estimate of variability, and an explicit analysis of how the available clinical variables contribute relative to the gene signal. This thesis is a step in that direction and opens room for further research on applying evolutionary algorithms to small, high-dimensional biomedical datasets.

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